WS01 Environmental and Water Chemistry Data

These datasets contain environmental monitoring and stream chemistry data associated with Watershed 1 (WS01) at the H.J. Andrews Experimental Forest. Environmental data include discharge and measurements from the WS01 gage and nearby PRIMET climate station, while water chemistry data are from the H.J. Andrews long-term stream chemistry monitoring program. Some environmental variables are subsequently incorporated into the processed dissolved oxygen (DO) dataset, which is documented separately.

1. HJA_climate_clean.csv

Description: Continuous environmental monitoring data compiled from the WS01 gage and PRIMET climate station for 2024–2026. Annual datasets were combined by timestamp. Discharge was calculated from stream stage using the applicable WS01 rating curve and converted to liters per second. Flagged water-temperature and electrical-conductivity measurements were removed during processing.

A subset of these variables is subsequently incorporated into the processed DO dataset. Specifically, gage electrical conductivity, discharge, total precipitation, and accumulated precipitation are aligned to 15-minute intervals and joined to the DO sensor observations. Pasted text

Unit of observation: Each row represents an environmental observation at a specified date and time. Source measurements are primarily recorded at 5-minute intervals.

Variables:

- Date – Date and time of observation.
- STAGE_INST_0_0_01 – Instantaneous stream stage at the WS01 gage (ft).
- Discharge_cfs – Stream discharge calculated from stage using the WS01 rating curve (ft^3/s).
- Discharge_L_s – Stream discharge calculated from stage using the WS01 rating curve (L/s).
- ECD_INST_0_0_01 – Electrical conductivity measured at the WS01 gage (µS/cm).
- WATERTEMP_MEAN_0_0_01 – Mean stream water temperature (°C).
- AIRTEMP_MEAN – Mean air temperature (°C).
- AIRTEMP_MAX – Maximum air temperature (°C).
- AIRTEMP_MIN – Minimum air temperature (°C).
- PRECIP_TOT_100_0_01 – Precipitation during the measurement interval (mm).
- PRECIP_ACC_100_0_01 – Accumulated precipitation (mm).
- Additional variables – Additional environmental measurements retained from the original WS01 gage and PRIMET datasets.

Use in DO dataset: The processed DO dataset includes selected variables from this file as `Gage_EC_us_cm`, `Discharge_L_s`, `Gage_precip_total_mm`, and `Gage_precip_accumulate_mm`.

Pasted text These are joined to DO observations after timestamps are rounded to 15-minute intervals. The final DO dataset is maintained and documented separately.

2. HJA_nutrient.csv

Description: Stream chemistry measurements from the H.J. Andrews long-term stream-water chemistry monitoring program for WS01. All chemistry observations in this dataset represent **3-week composite samples** collected using a streamflow-proportional sampling approach. Records from the 2024 and 2025 source datasets were standardized and combined for use with the WS01 DO study.

Unit of observation: Each row represents one **3-week composite stream-water chemistry sample** from WS01. The `date_time` variable identifies the date and time associated with the composite sample.

Variables:

- date_time – Date and time associated with the 3-week composite sample.
- ph – Stream-water pH (pH units).
- alk – Alkalinity expressed as calcium carbonate ($CaCO_3$) (mg/L).
- ssed – Suspended sediment concentration (mg/L).
- si – Dissolved silica concentration (mg/L).
- utp – Unfiltered total phosphorus concentration (mg/L).
- tdp – Total dissolved phosphorus concentration (mg/L).
- partp – Particulate phosphorus concentration (mg/L).
- po4p – Orthophosphate phosphorus concentration (mg/L).
- utn – Unfiltered total nitrogen concentration (mg/L).
- tdn – Total dissolved nitrogen concentration (mg/L).
- don – Dissolved organic nitrogen concentration (mg/L).
- partn – Particulate nitrogen concentration (mg/L).
- nh3n – Ammonia/ammonium nitrogen concentration (mg/L).
- no3n – Nitrate nitrogen concentration (mg/L).
- na – Dissolved sodium concentration (mg/L).
- k – Dissolved potassium concentration (mg/L).
- ca – Dissolved calcium concentration (mg/L).
- mg – Dissolved magnesium concentration (mg/L).
- so4s – Dissolved sulfate expressed as sulfur (SO_4-S) (mg/L).
- cl – Dissolved chloride concentration (mg/L).
- doc – Dissolved organic carbon concentration (mg/L).

Relationship to DO Data: HJA_nutrient.csv is maintained separately from the high-frequency DO dataset because each chemistry observation represents a 3-week composite rather than an instantaneous measurement. For analyses relating chemistry to DO, DO conditions can therefore be summarized over the corresponding 3-week sampling period rather than matched to a single sampling timestamp.